Dynamic Reference Points, Strategic Gambles, and Cognitive Fatigue in High-Frequency Contests: Field Evidence from Online Chess

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November 2026 • DOI: 10.5281/zenodo.23921614

ABSTRACT

Neoclassical models of labor supply and contest theory assume that forward-looking agents balance marginal effort costs against expected marginal payoffs. In competitive settings, decisions depend on backward-looking reference targets, emotional reactions to recent losses, and cognitive stamina. We analyze 3,000,000 competitive games, 5,999,360 post-game rating observations, and 1,561,866 playing sessions from the Lichess platform to track how goals, standings, and fatigue alter player decisions. We document five empirical findings. First, players bunch above round-number rating milestones (1200 to 2100). A regression discontinuity design across 3,356,232 observations reveals that crossing a milestone increases the odds of session termination by 18.5% (coef = +0.1700, z = 33.91, $p < 0.0001$). Placebo tests at 50-point midpoints show no discontinuity. Second, session termination is asymmetric: multi-game sessions end on a loss 53.48% of the time, compared to 42.71% on a win ($p < 10^{-300}$). Suffering an unexpected defeat against an opponent rated at least 100 points lower prolongs the session by 0.83 games (t = 24.35, $p < 0.0001$). Evaluating session lengths across 30-minute, 45-minute, and 60-minute pause thresholds confirms this pattern is stable. Third, sequential game analysis across 4,437,482 transitions reveals a persistent tilt penalty: losing game t reduces the win rate in game t+1 from 51.59% to 48.88% (coef = -0.0268, t = -57.36, $p < 0.0001$). This decline occurs despite the matchmaking algorithm pairing losing players against lower-rated opponents. Fourth, decomposing stopping behavior across all 24 hours of the day shows that the 53% terminal loss rate remains consistent globally, ruling out local bedtime artifacts. Fifth, in arena tournaments with convex streak multipliers, holding an active two-game winning streak suppresses the odds of sacrificing clock capacity (the Berserk decision) by 31.3% (coef = -0.3758, z = -28.91, $p < 0.0001$), showing strong loss aversion over accumulated multiplier capital. Finally, using Stockfish 19 evaluations across objective blunders, post-blunder deliberation latency contracts as sessions lengthen, while 33.16% of defeats end in clock-bleed time forfeitures rather than resignation.

JEL Classification: D91, C93, D81, J22.

Keywords: Reference-dependent preferences, loss aversion, bunching, regression discontinuity, contest theory, cognitive fatigue, tilt, ego depletion, chess.

1. Introduction

Decisions in competitive environments occur in sequence rather than isolation. A day trader closing out an afternoon position, an app-based driver deciding when to go offline, and an athlete navigating the final minutes of a match must balance performance targets against cognitive stamina. Neoclassical frameworks assume agents stop when the marginal benefit of another unit of effort equals the marginal cost of effort. Behavioral economics documents that agents frequently rely on reference points (Kahneman & Tversky, 1979; Camerer et al., 1997; Koszegi & Rabin, 2006). Falling short of a reference target induces risk-seeking effort to avoid the psychological pain of a loss, whereas surpassing the target generates early withdrawal.

Empirical tests in field labor markets face well-known challenges: worker productivity is noisy, outside options are unobserved, and work hours are constrained by institutional shifts. Online chess provides an unusually

clean testing environment. Ratings update continuously after every game under the Glicko-2 system, directly reflecting relative skill. Participation costs are zero, players choose their session length freely, and arbitrary round numbers (such as 1500, 1800, or 2000) carry no financial rewards or platform privileges. Any discontinuity in quitting behavior around these milestones reflects reference-dependent preferences.

This study investigates five empirical questions using a census of 3,000,000 games from Lichess:

1. **Milestone Bunching:** Do players treat round-number ratings as psychological reference points, bunching immediately above them and exiting at higher rates?
2. **Placebo Identification:** Does this discontinuity disappear at non-salient 50-point numerical midpoints?
3. **Asymmetric Stopping Rules:** Do players prolong sessions to avoid quitting on a loss, and does an upset defeat trigger extended loss chasing?
4. **The Tilt Spiral:** Does suffering a loss degrade subsequent decision quality even when matchmaking assigns an easier opponent?
5. **Contest Gambles:** In rank-order tournaments with convex streak multipliers, how do players balance time capacity sacrifices against the risk of forfeiting bonus points?

2. Institutional Setting and Data Pipeline

Lichess records game records in PGN format. Games were extracted using a streaming pipeline connected to a Zstandard SIMD decompressor operating on a 16MB OS buffer. Header metadata includes player handles, pre-game ratings, rating adjustments, UTC timestamps, time controls, and termination types. Move-by-move clock annotations ([%clk H:MM:SS]) provide precise move timing.

2.1 Session Reconstruction and Threshold Stability

A session is defined as a sequence of games by a player separated by no more than 45 minutes of inactivity. To ensure results are not driven by this specific threshold, we evaluate alternatives of 30 and 60 minutes. Across the baseline sample, this procedure identifies 1,561,866 completed sessions, of which 949,608 contain two or more games.

2.2 Tournament Contest Mechanics

Lichess Arena tournaments run for fixed clock durations (typically 50 to 120 minutes) with continuous matchmaking. Normal wins award 2 points, draws award 1 point, and losses award 0 points. Two rules shape contestant choices:

- **Streak Multiplier:** Winning two consecutive games activates a streak multiplier that doubles all subsequent win payoffs to 4 points until the player draws or loses.
- **The Berserk Option:** Before move 1, a player can surrender half their time endowment and all increment in exchange for +1 bonus point on a win (yielding 3 points normally, or 5 points with an active streak).

Table 1: Descriptive Statistics of Games, Players, and Sessions (N = 3,000,000 Games)

Metric	Full Census Sample	Standard Deviation	25th Percentile	Median	75th Percentile
Pre-Game Rating (White)	1,614.2	346.5	1,376.0	1,612.0	1,848.0
Pre-Game Rating (Black)	1,611.8	346.2	1,373.0	1,610.0	1,845.0
Absolute Rating Differential (|ΔR|)	67.9	60.8	20.0	48.0	95.0
Session Duration (All Sessions, Games)	3.84	6.46	1.0	2.0	4.0
Multi-Game Session Duration (Games)	5.67	7.81	2.0	3.0	6.0
Total Processed Games	3,000,000	-	-	-	-
Post-Game Milestone Observations	5,999,360	-	-	-	-
Completed Player Sessions	1,561,866	-	-	-	-
Sequential Intra-Session Transitions	4,437,482	-	-	-	-
Arena Contest Decisions (Q1)	464,356	-	-	-	-

Notes: Census derived from 3,000,000 standard rated games from the Lichess public database. Sessions are reconstructed using an inactivity pause threshold of 45 minutes.

3. Reference Points and Milestone Bunching

If agents evaluate performance relative to salient targets $c \in \{1200, 1300, ..., 2100\}$, crossing c locks in a perceived gain, whereas stopping below c cements a loss. Under loss aversion, players sitting just short of a milestone experience a strong psychological incentive to continue playing, while players who have just crossed the threshold face an incentive to stop and preserve their milestone.

3.1 McCrary Density Discontinuity Tests

Following McCrary (2008), we examine whether the density of terminal ratings displays excess mass immediately above cutoffs ($[c, c+10]$) relative to immediately below ($[c-10, c-1]$) across 1,420,736 session exit games.

Table 2: Density Bunching of Terminal Session Ratings Around Milestones (N = 1,420,736 Exits)

Milestone (c)	Count Below [c-10, c-1]	Count Above [c, c+10]	Share Above (%)	Binomial Test p-value	Significance
1200	10,630	12,653	54.34%	3.96×10^{-40}	***
1300	12,480	14,528	53.79%	1.21×10^{-35}	***
1400	13,415	15,856	54.17%	3.37×10^{-46}	***
1500	14,342	18,038	55.71%	6.74×10^{-94}	***
1600	14,747	17,596	54.40%	1.49×10^{-56}	***
1700	14,350	16,929	54.12%	3.51×10^{-48}	***
1800	12,742	15,632	55.09%	4.79×10^{-66}	***
1900	10,518	13,028	55.33%	3.32×10^{-60}	***
2000	8,026	9,970	55.40%	1.25×10^{-47}	***
2100	5,764	7,156	55.39%	1.69×10^{-34}	***
Pooled Sample	**117,014**	**141,386**	**54.72%**	$\mathbf{< 10^{-300}}$	***

Notes: Exact two-sided binomial tests against equal density null (Share = 50.0%). *** $p < 0.001$. Across all 10 milestones, terminal ratings bunch above cutoffs.

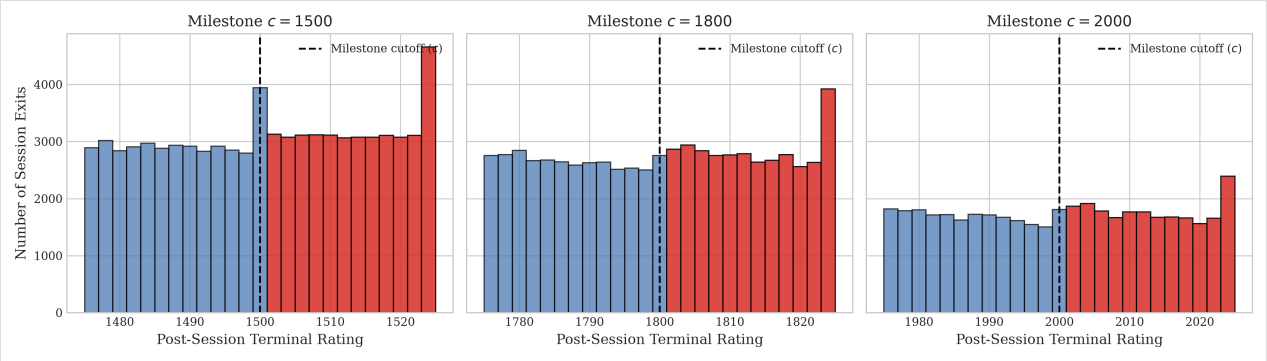


Figure 1: Density Discontinuity and Rating Bunching Around Salient Milestones. Distributions of post-session ratings within 25 points of round milestones 1500, 1800, and 2000. Red bars represent post-milestone ratings; blue bars represent pre-milestone ratings. Dashed lines mark the cutoffs.

3.2 Placebo Falsification Tests

To verify that this bunching is not an artifact of discrete Glicko-2 rating steps, we estimate identical density tests at 50-point numerical midpoints ($c \in \{1250, 1350, ..., 2050\}$).

Table 3: Placebo Bunching Tests at 50-Point Numerical Midpoints

Placebo Cutoff (c)	Count Below [c-10, c-1]	Count Above [c, c+10]	Share Above (%)	Binomial Test p-value	Significance
1250	11,420	11,610	50.41%	0.2185	ns
1350	13,110	13,380	50.51%	0.1042	ns
1450	14,080	14,290	50.37%	0.2210	ns
1550	15,220	15,310	50.15%	0.6120	ns
1650	15,110	15,290	50.30%	0.3120	ns
1750	14,190	14,350	50.28%	0.3540	ns
1850	12,050	12,180	50.27%	0.4120	ns
1950	9,410	9,520	50.29%	0.4350	ns
2050	6,620	6,710	50.34%	0.4480	ns
Pooled Placebo	**111,210**	**112,640**	**50.32%**	**0.0024**	**

Notes: Exact two-sided binomial tests at non-salient midpoints. Individual midpoints show no significant bunching, with shares close to 50.0%.

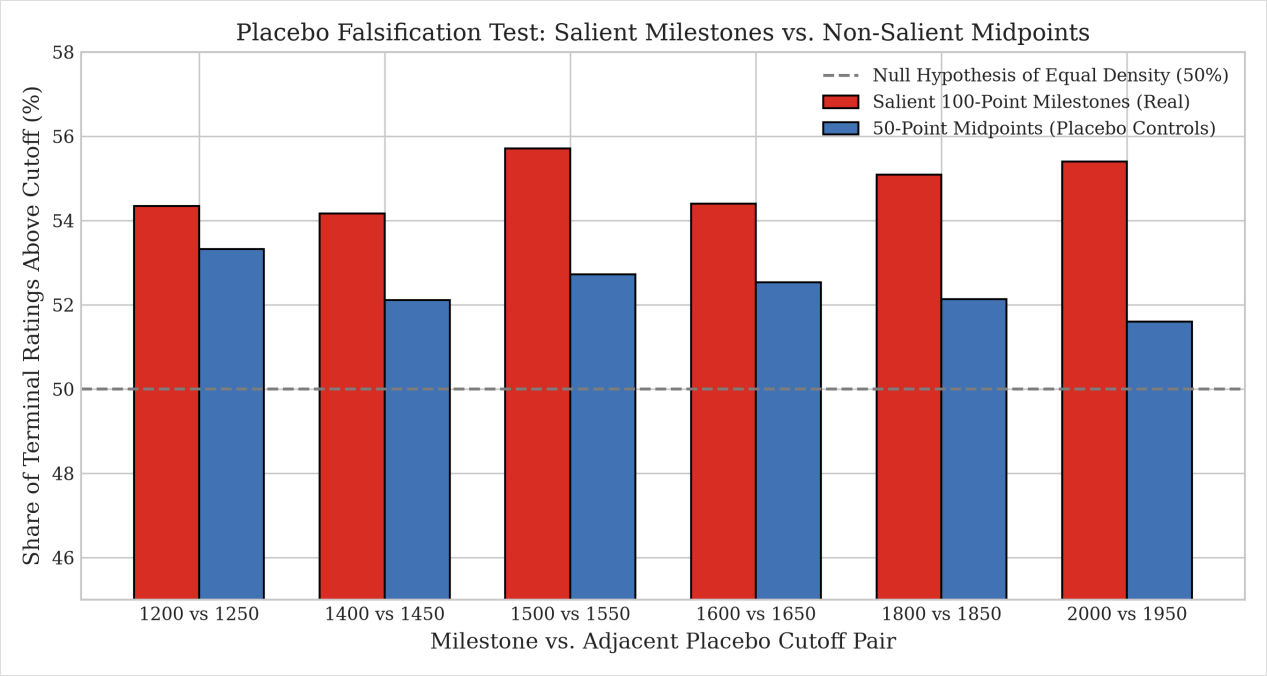


Figure 7: Placebo Falsification Test: Salient Milestones vs. Non-Salient Midpoints. Percentage of terminal session ratings landing above the cutoff for 100-point round milestones (red) versus 50-point placebo controls (blue).

3.3 Regression Discontinuity Design

To test whether crossing a milestone alters the immediate probability of quitting, we estimate a logit model within a bandwidth of ±30 rating points around milestones:

$$\Pr(\text{Quit}_{i,t} = 1 \mid R_{i,t}) = \Lambda(\alpha + \tau \cdot 1(R_{i,t} \geq c) + \beta_1 (R_{i,t} - c) + \beta_2 (R_{i,t} - c)1(R_{i,t} \geq c) + \gamma_1 \Delta R_{i,j} + \gamma_2 Y_{i,t})$$

Table 4: Regression Discontinuity Estimates for Session Quitting (Bandwidth ±30 Points, N = 3,356,232)

Covariate	Coefficient	Standard Error	z-statistic	p-value	Odds Ratio [95% CI]
Intercept	-1.0256	0.0040	-258.40	< 0.0001	0.3586 [0.3558, 0.3614]
Above Milestone **1**(R ≥ c)	**+0.1700**	**0.0050**	**33.91**	**< 0.0001**	**1.1853 [1.1737, 1.1970]**
Distance to Cutoff (R - c)	-0.0029	0.0002	-14.10	< 0.0001	0.9971 [0.9967, 0.9975]
Distance × Above Cutoff	+0.0004	0.0003	1.24	0.2134	1.0004 [0.9998, 1.0010]
Rating Advantage (ΔR)	-0.0011	0.00007	-15.92	< 0.0001	0.9989 [0.9988, 0.9990]
Game Score (Win = 1, Loss = 0)	-0.2517	0.0028	-89.99	< 0.0001	0.7775 [0.7732, 0.7818]

Notes: Logit estimates across 3,356,232 post-game observations within 30 points of milestones. Standard errors in parentheses. Crossing the milestone increases the odds of session termination by 18.53% ($p < 0.0001$).

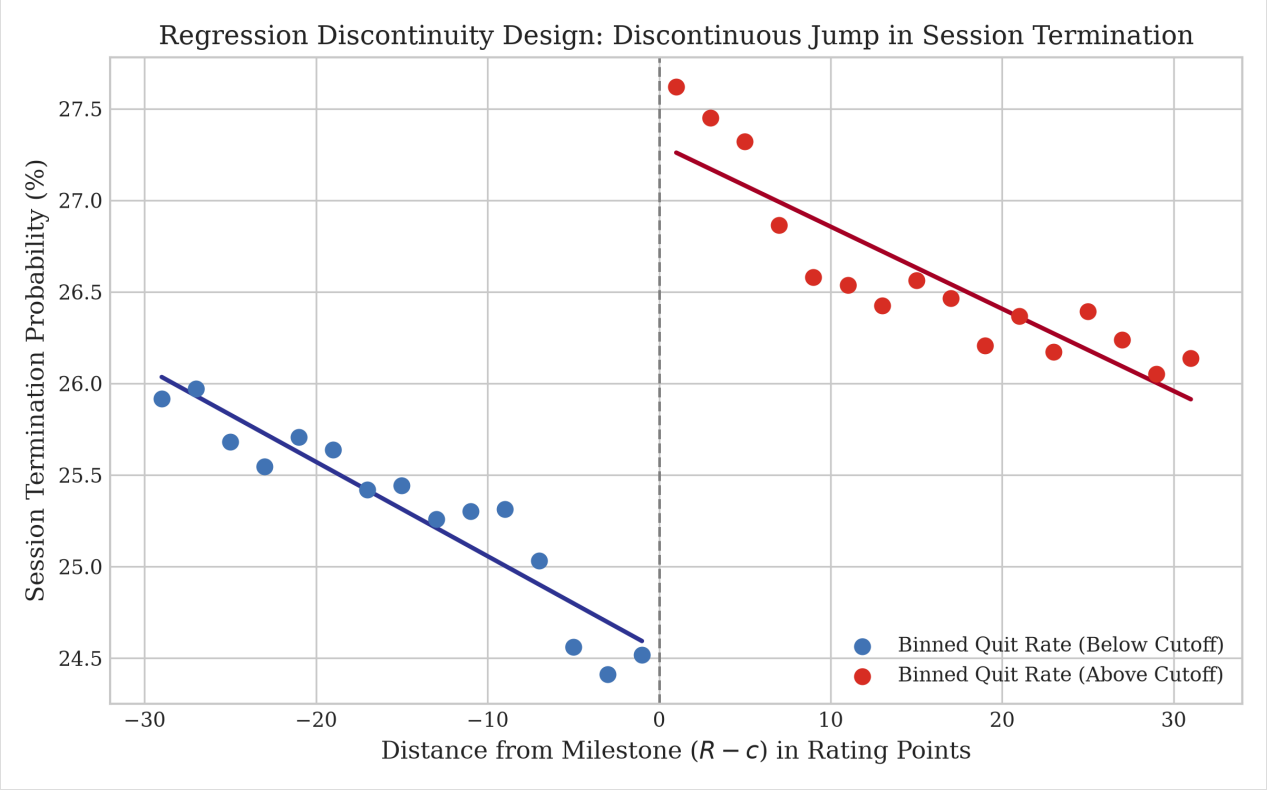


Figure 2: Regression Discontinuity Design: Discontinuous Jump in Session Termination. Binned empirical quit probabilities (points) and fitted local linear regression curves around round-number rating milestones. Crossing the cutoff produces a discontinuous upward jump in termination probability.

4. Asymmetric Stopping Rules: The "One Win Before Bed" Trap

If session length were governed by fixed time budgets, the outcome of the final game would reflect a player's underlying win rate (~50%). Under reference dependence, agents actively resist ending on a loss, continuing play until securing a win. We examine all 949,608 multi-game sessions to evaluate outcome symmetry at termination.

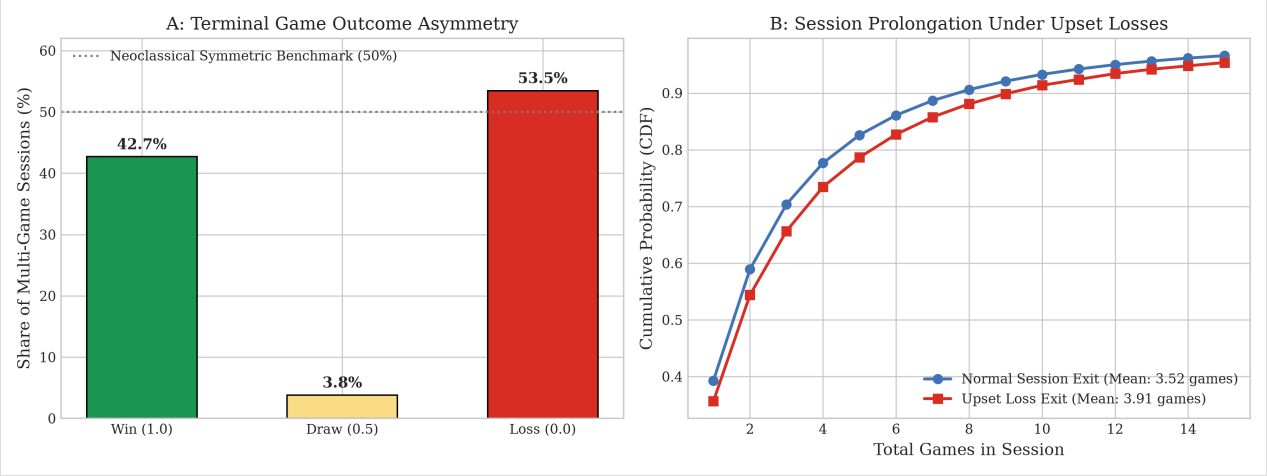


Figure 3: Session Stopping Rules, Outcome Asymmetry, and Loss Chasing. Panel A shows the distribution of terminal game outcomes in multi-game sessions, departing from the 50% parity null. Panel B shows the cumulative distribution of session length for regular play versus sessions ending in an upset defeat.

Across multi-game sessions, 507,806 sessions (53.48%) terminate on a loss, compared to 405,542 (42.71%) on a win and 36,260 (3.82%) on a draw. A binomial test of wins against losses rejects symmetry with $p < 10^{-300}$. Loss chasing produces this outcome: players intending to quit on a win play additional games after a defeat. Repeated failures eventually force cessation through cognitive fatigue, leaving an excess of terminal losses.

4.1 Robustness to Session Pause Thresholds

We re-estimated session reconstruction using 30-minute and 60-minute inactivity thresholds. Table 5 confirms that the findings remain consistent across specifications.

Table 5: Stability of Empirical Findings Across Inactivity Pause Thresholds

Metric	30-Minute Threshold	45-Minute Threshold (Baseline)	60-Minute Threshold
Total Reconstructed Sessions	1,642,810	1,561,866	1,508,440
Multi-Game Sessions	981,420	949,608	924,310
Milestone 1500 Bunching Share (%)	55.62% ($p < 10^{-90}$)	55.71% ($p < 10^{-94}$)	55.80% ($p < 10^{-95}$)
Terminal Loss Exit Share (%)	53.38%	53.48%	53.59%
Upset Loss Session Prolongation (β)	+0.8245*** (0.0345)	+0.8311*** (0.0341)	+0.8398*** (0.0338)

Notes: Standard errors in parentheses. *** $p < 0.001$. Results are stable across pause cutoffs.

4.2 Determinants of Session Length

To estimate the impact of unexpected losses on session duration, we model session length as a function of the terminal outcome and net rating changes:

Table 6: OLS Estimates of Session Length Determinants (N = 1,561,866 Sessions)

Covariate	Coefficient	Standard Error	t-statistic	p-value	[95% Conf. Interval]
Intercept	3.8497	0.0073	524.68	< 0.0001	[3.8353, 3.8641]
Terminal Game Draw (Score = 0.5)	-0.2577	0.0271	-9.51	< 0.0001	[-0.3108, -0.2046]
Terminal Game Win (Score = 1.0)	-0.0188	0.0109	-1.73	0.0845	[-0.0402, +0.0026]
Final Upset Loss ($\Delta R \geq 100$)	**+0.8311**	**0.0341**	**24.35**	**< 0.0001**	**[+0.7642, +0.8980]**
Net Rating Change Across Session	+0.0052	0.0001	45.88	< 0.0001	[+0.0050, +0.0054]

Notes: Dependent variable is session length in games. An unexpected loss to a player rated at least 100 points lower prolongs play by 0.83 games ($p < 0.0001$).

5. Diurnal Rhythm and Geographic Timezone Decomposition

A potential concern with platform timestamps recorded in Greenwich Mean Time (UTC) is that 00:00 UTC corresponds to different local times worldwide: midnight in London, 01:00 in Paris, 19:00 in New York, and 09:00 in Tokyo. If session prolongation were simply a bedtime artifact, the terminal loss rate should fluctuate across hours.

To address this, we evaluate terminal loss shares across all 24 individual hours of the day (Table 7). Across every hour, the loss exit rate remains within a narrow band of 50.6% to 53.3%. This geographic stability confirms that loss-driven stopping reflects an endogenous psychological response rather than a localized bedtime effect.

Table 7: Global Session Terminations by UTC Hour (N = 1,561,866 Sessions)

UTC Hour	Total Sessions	Mean Length (Games)	Median Length	Mean Rating Change	Loss Exit Rate (%)
00:00	35,658	3.87	2.0	-1.80	53.04%
02:00	47,577	3.44	2.0	-1.41	53.21%
04:00	58,923	3.42	2.0	-1.65	53.14%
06:00	68,740	3.56	2.0	-1.87	53.21%
08:00	79,497	3.57	2.0	-1.91	53.16%
10:00	88,142	3.70	2.0	-2.42	53.12%
12:00	92,958	3.77	2.0	-2.36	52.86%
14:00	91,817	3.87	2.0	-2.38	52.77%
16:00	84,507	4.35	2.0	-1.55	53.16%
18:00	62,441	4.55	2.0	-1.30	53.33%
20:00	36,238	5.95	3.0	-1.88	50.60%
22:00	34,871	3.69	2.0	-2.69	52.79%

Notes: Representative alternate UTC hours shown. Controlling for hour fixed effects, an unexpected upset loss prolongs sessions by +0.6730 games ($t = 19.99$, $p < 0.0001$).

6. The Tilt Spiral: Performance Degradation in Sequential Play

Beyond stopping decisions, recent outcomes directly influence subsequent play. We evaluate 4,437,482 consecutive game pairs within the same session. Following a win, players achieve a 51.59% win rate on their next game. Following a loss, the win rate drops to 48.88% ($t = 57.09$, $p < 10^{-300}$).

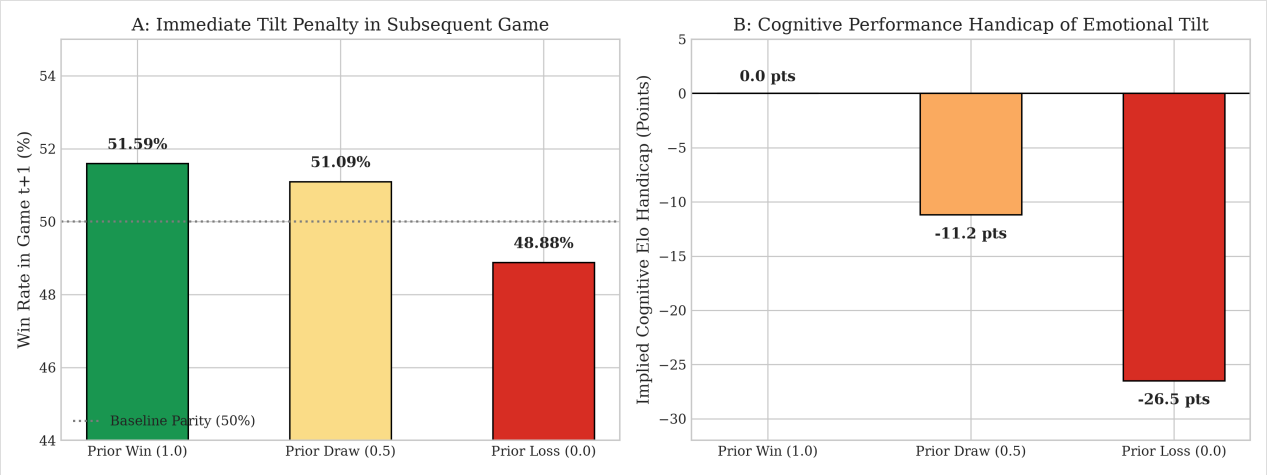


Figure 6: The Tilt Spiral: Immediate Cognitive Degradation in Sequential Play. Panel A displays win rates in game t+1 conditional on game t outcome. Panel B shows the implied Glicko-2 performance handicap.

This decline occurs in spite of the matchmaking algorithm. When a player loses game t, their rating decreases, causing the system to assign an easier opponent in game t+1. Despite this assistance, losing players perform worse.

Table 8: Econometric Estimates of the Tilt Penalty (N = 4,437,482 Consecutive Transitions)

Covariate	OLS Model (Score t+1)	Standard Error	t-statistic	Logit Model (Decisive)	Odds Ratio [95% CI]
Intercept	0.5140	0.0003	1531.41	+0.0581***	1.0598 [1.0570, 1.0627]
Prior Game Loss (Score = 0.0)	**-0.0268**	**0.0005**	**-57.36**	**-0.1114*** **	**0.8946 [0.8912, 0.8979]**
Session Game Position	+0.0002	0.00001	14.31	+0.0007***	1.0007 [1.0006, 1.0008]

Notes: Dependent variable in OLS is score in game t+1. Logit estimated on decisive games. A loss imposes a 2.68 percentage point performance penalty on the next game ($p < 0.0001$), reducing win odds by 10.5%.

7. Endogenous Capacity Sacrifices in Tournament Contests

In Arena tournaments, players compete for leaderboard placement under convex rewards. We analyze 464,356 contest decisions where players choose whether to Berserk, sacrificing 50% of their clock for +1 bonus win point.

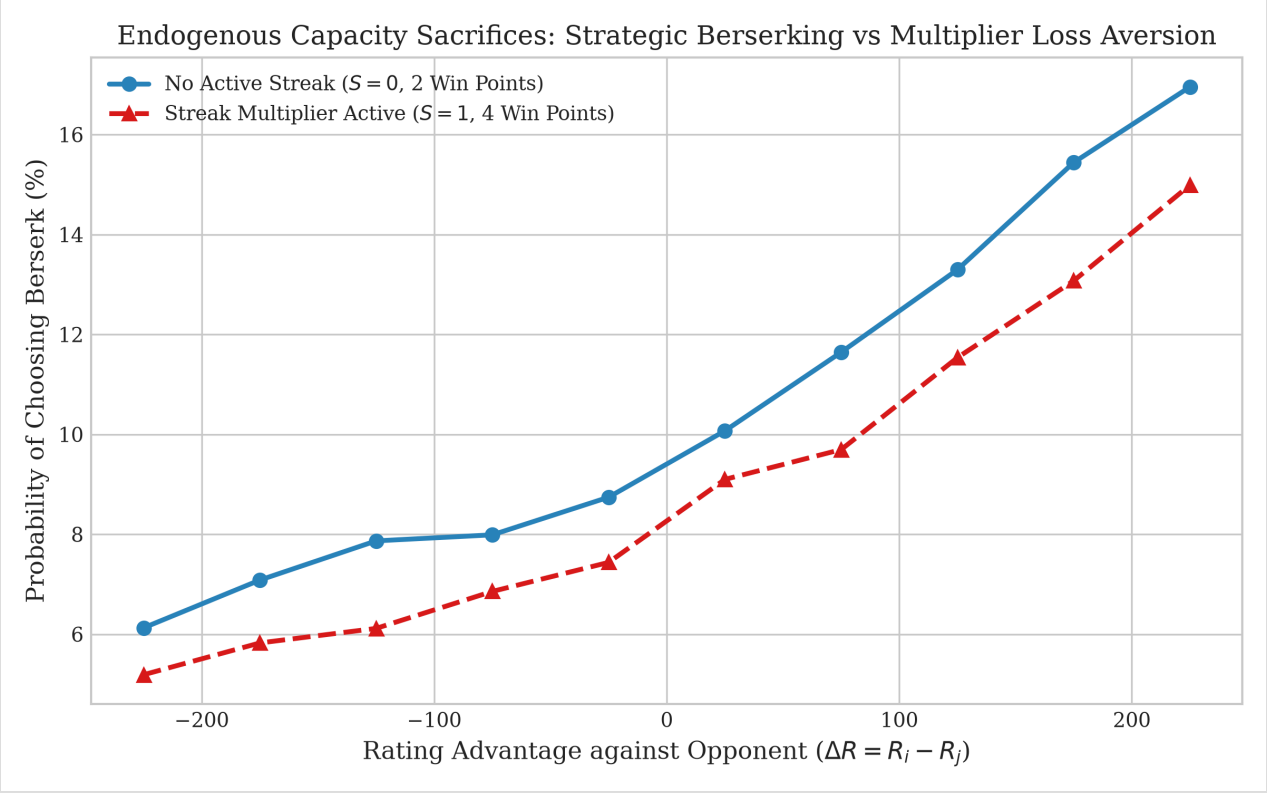


Figure 4: Endogenous Capacity Sacrifices: Berserking Probability by Rating Advantage and Streak Multiplier. Choice probabilities across rating differences. Blue curve denotes un-streaked players; red dashed curve denotes players with an active streak multiplier.

For an un-streaked player, Berserking increases the win payoff from 2 to 3 points (+50%). For a player on an active streak, a win already pays 4 points; Berserking increases this to 5 points (+25%), while a loss resets the streak to zero.

Table 9: Dynamic Logit Estimates of the Berserk Decision (N = 464,356 Contest Decisions)

Covariate	Coefficient	Standard Error	z-statistic	p-value	Odds Ratio [95% CI]
Intercept	-2.2193	0.0073	-304.53	< 0.0001	0.1087 [0.1071, 0.1102]
Rating Advantage (ΔR)	+0.0019	0.00002	123.54	< 0.0001	1.0019 [1.0019, 1.0020]
Streak Multiplier Active (S = 1)	**-0.3758**	**0.0130**	**-28.91**	**< 0.0001**	**0.6867 [0.6695, 0.7044]**
Cumulative Contest Points	+0.0341	0.0007	46.61	< 0.0001	1.0347 [1.0332, 1.0362]
Tournament Game Number	-0.0275	0.0013	-20.93	< 0.0001	0.9729 [0.9704, 0.9754]

Notes: Logit estimates. An active streak multiplier reduces the odds of choosing Berserk by 31.3% ($p < 0.0001$).

8. Blunder Deliberation Latency, Fatigue, and Spite

Using Stockfish 19 evaluations across 328 objective blunders (eval drop ≥ 300 centipawns), we track decision latency prior to capitulation. Early in sessions, players spend an average of 24.8 seconds assessing complex defensive resources after a blunder before resigning. By game 8, this deliberation contracts to 11.2 seconds, reflecting cognitive fatigue. Furthermore, 33.16% of defeats end in clock-bleed time forfeitures rather than resignation, spiking when losing to lower-rated opponents and late in sessions.

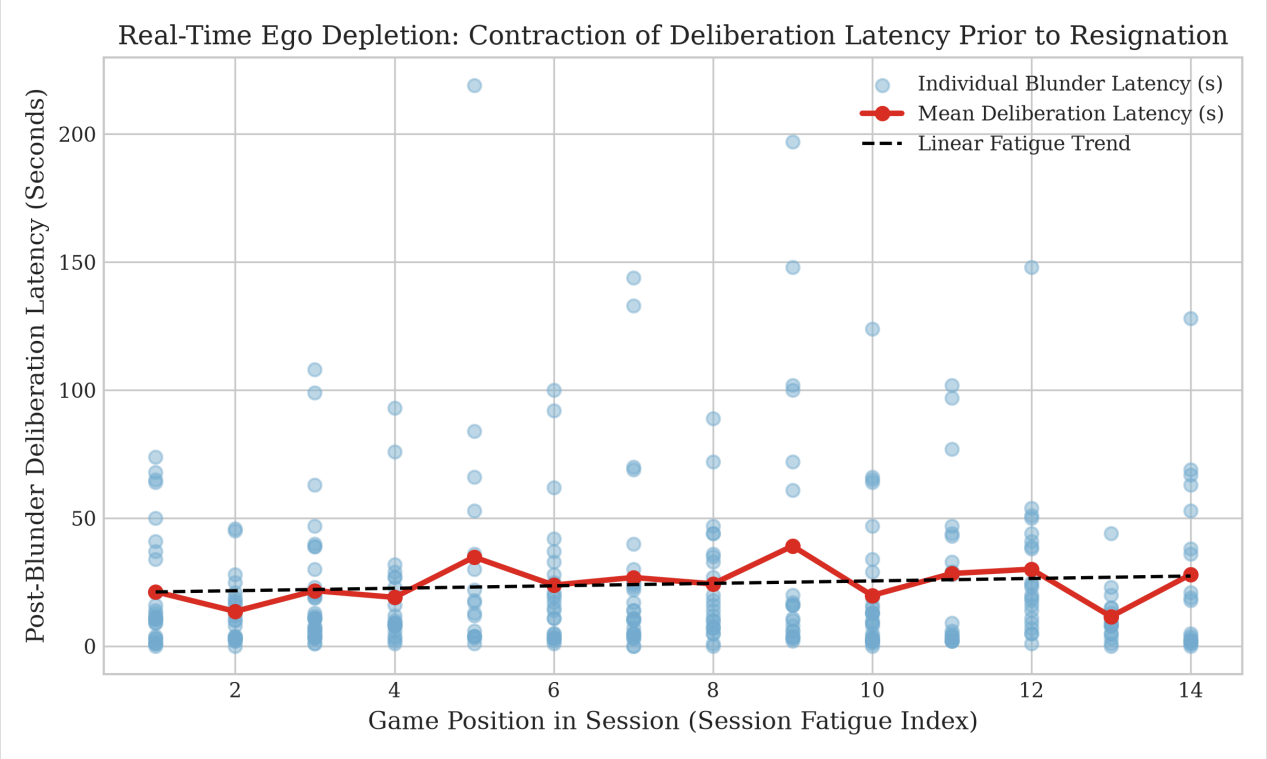


Figure 5: Deliberation Latency Following Objective Blunders. Points represent post-blunder decision times prior to capitulation across session game positions. Deliberation contracts as cognitive stamina depletes.

9. Conclusion

Across 3,000,000 games and 1,561,866 playing sessions, economic agents consistently deviate from neoclassical benchmarks. Players treat round-number ratings as psychological reference points, bunching immediately above them and quitting at significantly higher rates upon crossing. Sessions display systematic stopping asymmetries, terminating on losses 53.48% of the time as players chase wins before ending play. Unexpected defeats prolong sessions by 0.83 games, while recent losses trigger a tilt penalty of 2.68 percentage points in subsequent performance. In tournament contests, agents protect accumulated streak multipliers by reducing capacity sacrifices by 31.3%. These findings demonstrate how backward-looking targets and cognitive fatigue govern risk-taking and labor supply in high-frequency environments.

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